Curiosity Rover Wheel Failure Analysis

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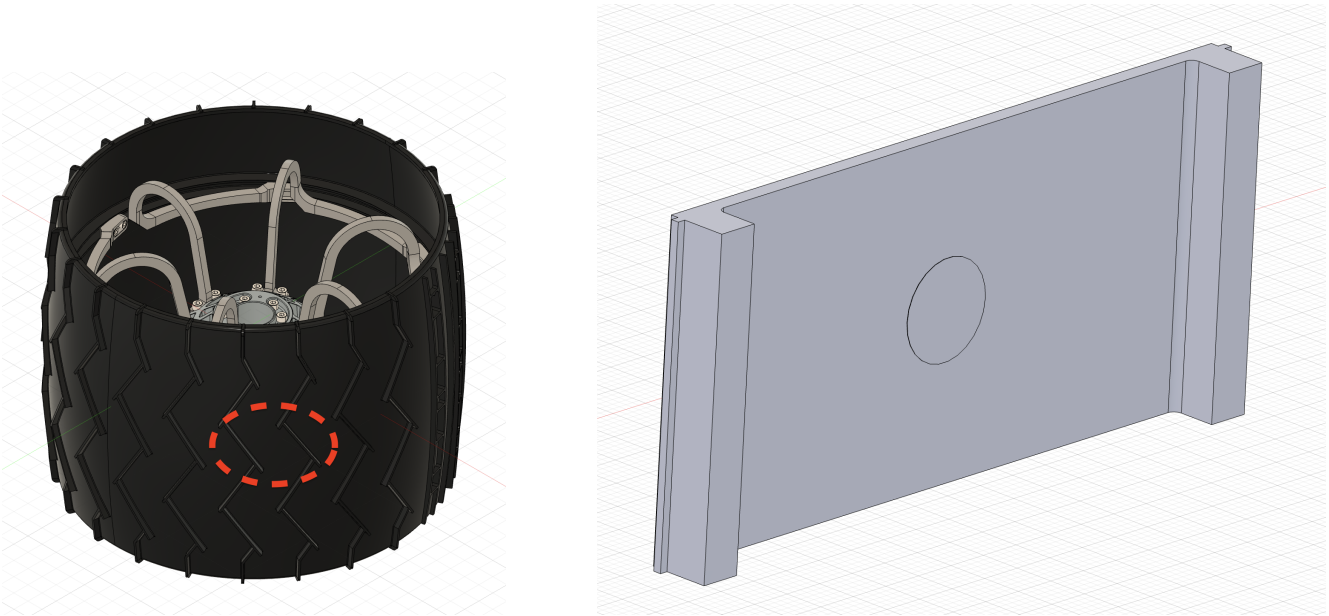
Abstract

The aluminium skin of the Curiosity rover wheel has cracked in service, with damage concentrated where the thin skin meets the grousers. This report examines a flat segment of that skin, spanning between two grousers and loaded by a rock pressing on a circular patch of radius a. A closed-form model, treating the skin as a fixed stepped beam with a Peterson stress concentration at the fillet, is compared with an Ansys 2025 R1 solid model along a path on the skin surface. The beam model reproduces the shape of the FEA stress distribution and the panel-centre peak for sharp contacts ($a = 2\,\text{mm}$, $C = 1.12$). It over-predicts the junction stress by a factor of three to six. For sharp rocks, the centre stress exceeds the 106 MPa fatigue limit of the material.

1 Introduction

NASA and JPL mission reports have documented punctures and cracks in the wheels of the Curiosity rover after driving over angular, embedded rocks [1]. The wheel skin is a thin aluminium shell stiffened by chevron-shaped grousers. A rock pressing on the skin between two grousers bends the skin like a plate clamped along its stiffened edges. The bending moment, and hence the surface stress, is largest at two places: directly under the rock, and at the skin–grouser junction, where the section changes abruptly and a small fillet concentrates the stress.

Tier 1 asks a narrow question: *for a given rock sharpness, how large are the junction and centre stresses, and can a hand calculation predict them?* The region studied is marked in Figure 1(a). It is idealised as the flat segment shown in Figure 1(b), and the patch radius a stands in for rock sharpness: a smaller a is a sharper rock. All CAD, data and scripts are in the project repository [3], and the report links to them rather than reproducing them.



(a) Region of the wheel analysed (b) Idealised skin segment with load patch

Figure 1: Location of the analysed segment on the wheel and the CAD model used.

2 Model and FEA setup

The segment is a 2 mm skin between two grouser blocks, each 5 mm wide and 7.5 mm deep overall. The skin meets each block through an R1 mm fillet (Table 1). The two outer end faces are fixed, which represents the continuation of the wheel shell beyond the grousers. The load is a uniform pressure on a circular patch at mid-span, with $a = 2, 4, 6, 8, 10$ mm. One CAD file per radius carries the patch as a split face.

The FEA was solved in Ansys Mechanical 2025 R1 with solid elements at a total force of 300 N. Stress was extracted as a *linearized equivalent stress* along a straight path 2 mm above the base, which is the top surface of the skin, running across the full 85 mm span (Figure 2). Ansys splits the stress along a path into membrane, bending and peak parts [2]. This report uses the *Total* column, which is the actual von Mises stress at each path point. The analysis load is a design choice of 400 N. The model is linear-elastic, so the FEA values are scaled by 400/300 before any comparison. The exported path data is in the repository.

Table 1: Model inputs.

Quantity	Symbol	Value
Skin thickness	t	2 mm
Grouser depth / width	D, w_g	7.5 mm, 5 mm
Overall length (fixed ends)	L	83 mm
Panel width	b	40 mm
Junction fillet radius	r	1 mm
Total load	P	400 N (FEA at 300 N, scaled)
Patch radius	a	2–10 mm
Material	–	Al 7075-T7351, $E = 72$ GPa, $\nu = 0.33$
Fatigue limit	σ_f	106 MPa

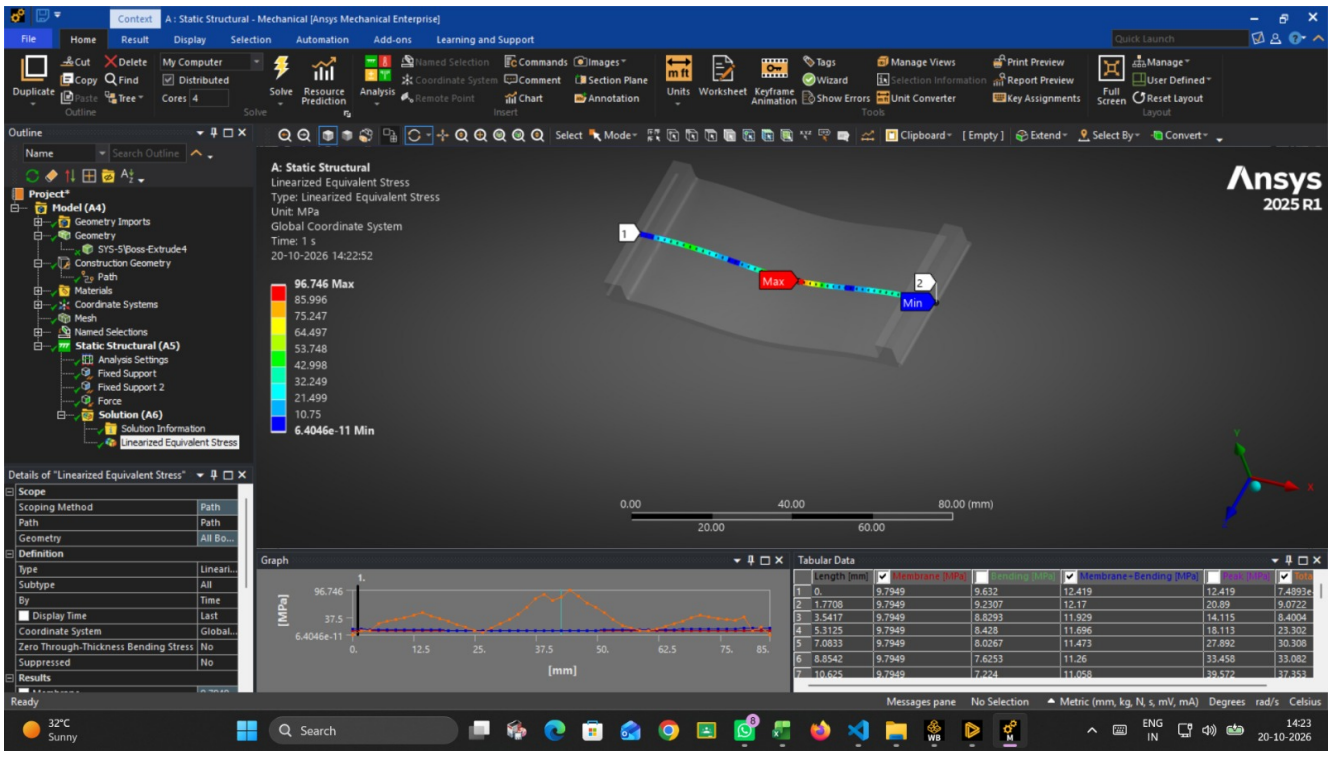


Figure 2: Ansys 2025 R1 run for $a = 2$ mm at 300 N: the linearized equivalent stress path on the skin, with the stress plotted along the path and the tabular output.

3 Analytical model

The model follows classical beam and plate theory in the form tabulated by Roark. It is implemented in `path_stress_analytical.m`, and the results are compared in `fea_vs_analytical.m`.

3.1 Patch load

A uniform pressure $P/\pi a^2$ on a circle of radius a, integrated across the width, gives a line load along the span x:

$$q(x) = \frac{P}{\pi a^2}\, 2\sqrt{a^2 - (x - x_c)^2}, \qquad |x - x_c| \le a, \tag{1}$$

which integrates to P. Here $x_c = L/2$ is the patch centre.

3.2 Fixed stepped beam

The segment is treated as a beam of width b, fixed at $x = 0$ and $x = L$. Its second moment of area $I(x)$ is $bD^3/12$ inside the grouser blocks and $bt^3/12$ over the skin. With $M_q(x)$ the moment of the load to the left of x,

$$M(x) = M_0 + V_0 x - M_q(x), \tag{2}$$

and the two unknown end reactions follow from zero slope and zero deflection at the far clamp:

$$\int_0^L \frac{M}{EI}\, dx = 0, \qquad \int_0^L \frac{M\,(L - x)}{EI}\, dx = 0. \tag{3}$$

This handles the stepped section and any load shape without tables. As $a \to 0$ with rigid grousers it recovers the textbook result $|M| = PL'/8$ at both the ends and the centre of the clear span L'. The script checks this limit on every run.

3.3 Stress on the path

On the skin surface the bending stress is $\sigma_x = 6M/(bt^2)$, the nominal stress σ_{nom}. At the junction the fillet raises it to

$$\sigma_{\max} = K_t\, \sigma_{\mathrm{nom}}, \tag{4}$$

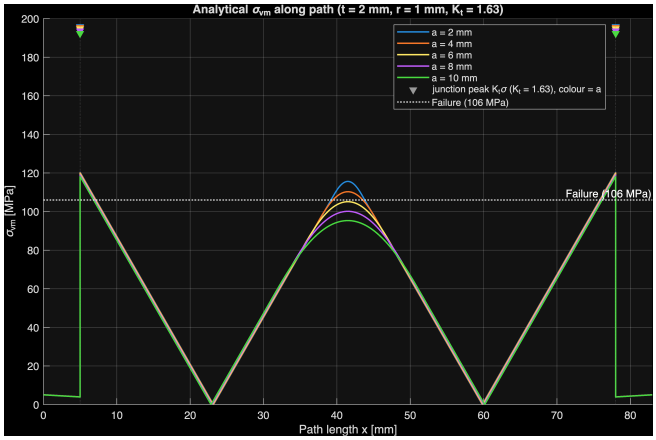
where K_t is taken from Peterson's fit for a stepped flat bar with a shoulder fillet in bending. With $h/r = 2.75$ and $2h/D = 0.73$, this gives $K_t = 1.63$. The top surface is traction-free ($\sigma_z = 0$), and a wide plate in cylindrical bending is restrained laterally, so $\sigma_y = \nu\sigma_x$. The principal stresses are then $(\sigma_x, \nu\sigma_x, 0)$, and the von Mises stress is

$$\sigma_{\mathrm{vm}} = |\sigma_x|\sqrt{1 - \nu + \nu^2}. \tag{5}$$

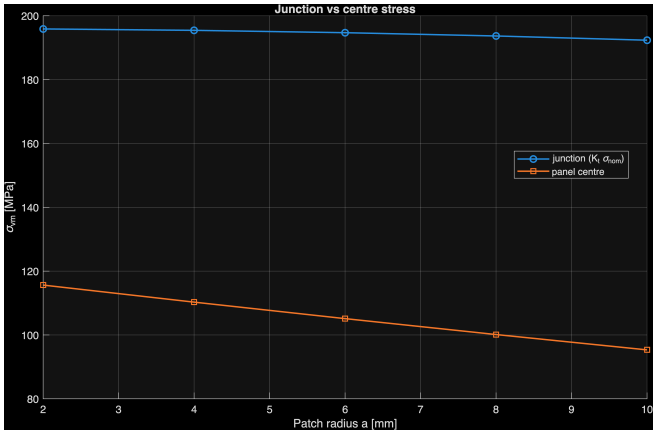
4 Results

4.1 Analytical

The analytical stress along the path (Figure 3a) is piecewise linear, as beam theory requires. It peaks at both junctions and under the patch, and passes through zero at the two inflection points near the quarter-spans. Patch size changes only the centre peak, because spreading the load flattens the moment under it, while the junction moment is almost insensitive to a (Table 2). The nominal junction stress of about 120 MPa becomes about 195 MPa once K_t is applied. Both values exceed the fatigue limit, as do the centre stresses for $a \le 6$ mm.



(a) σ_{vm} along the path



(b) Junction and centre peaks against a

Figure 3: Analytical results at 400 N. Triangles mark $K_t\sigma$ at the junctions; the dotted line is the 106 MPa fatigue limit.

Table 2: Analytical summary at 400 N (`summary.csv`).

a (mm)	M_j (N mm)	$\sigma_{\mathrm{nom},j}$ (MPa)	$K_t\sigma_{\mathrm{vm},j}$ (MPa)	$\sigma_{\mathrm{vm},c}$ (MPa)	ratio j/c
2	3636	136.4	195.9	115.6	1.69
4	3628	136.1	195.4	110.3	1.77
6	3614	135.5	194.7	105.1	1.85
8	3595	134.8	193.7	100.1	1.93
10	3571	133.9	192.3	95.3	2.02

4.2 FEA

The FEA stress along the path (Figure 4a) has the same three-peak shape, with minima close to the analytical inflection points. Away from the patch it also falls on the analytical slopes (Figure 5). The peaks were taken as the largest Total value within 15 mm of either end (junction) and within ±10 mm of mid-span (centre). The centre peak falls from 129 MPa at $a = 2$ mm to 42 MPa at $a = 10$ mm. The junction peaks stay between 30 and 65 MPa (Figure 4b).

NOTE: λ_j and λ_c represent data λ for the junction and the center, respectively

4.3 Comparison and calibration

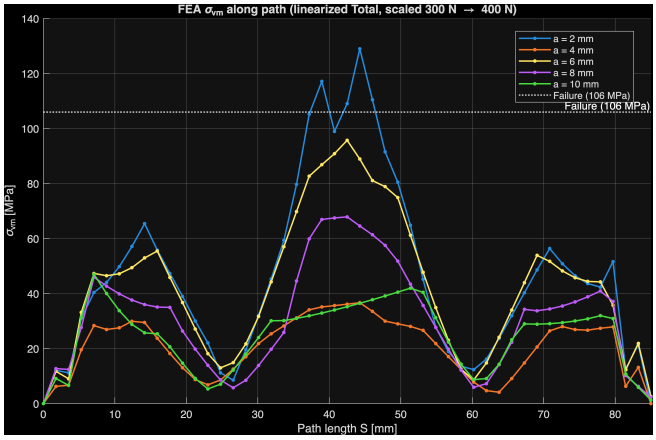
Figure 5 overlays the two solutions for each radius. The analytical span (83 mm) is centred on the 85 mm FEA path. The calibration factor $C = \sigma_{\mathrm{FEA}}/\sigma_{\mathrm{analytical}}$ is listed in Table 3.

Table 3: FEA against analytical peaks and calibration factors (`fea_vs_analytical.csv`). Stresses in MPa at 400 N.

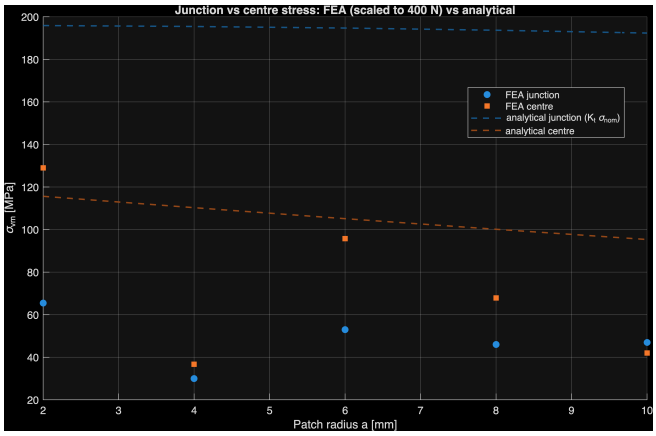
a (mm)	FEA j	Analytical j	C_j	FEA c	Analytical c	C_c
2	65.4	195.9	0.33	129.0	115.6	1.12
4	29.9	195.4	0.15	36.7	110.3	0.33
6	53.0	194.7	0.27	95.7	105.1	0.91
8	46.0	193.7	0.24	67.9	100.1	0.68
10	47.0	192.3	0.24	42.0	95.3	0.44

5 Discussion

Panel centre. For the sharpest contact the beam model is close: $C_c = 1.12$ at $a = 2$ mm, a 12% under-prediction. That is expected, because the beam spreads the load over the full 40 mm width while the real plate

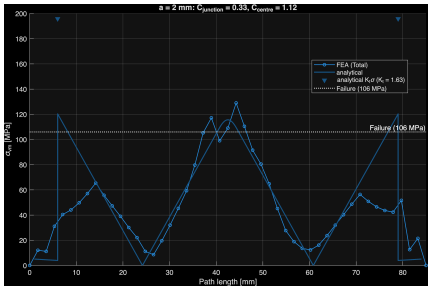


(a) FEA Total σ_{vm} along the path

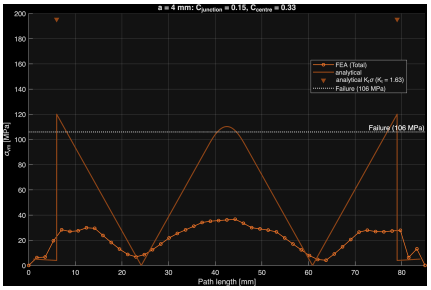


(b) FEA peaks (markers) against analytical (dashed)

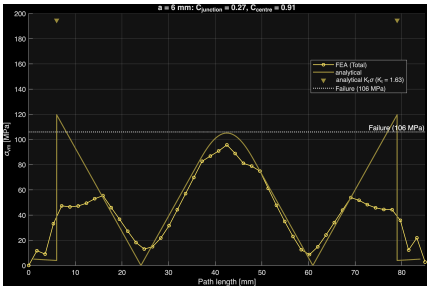
Figure 4: FEA results scaled to 400 N.



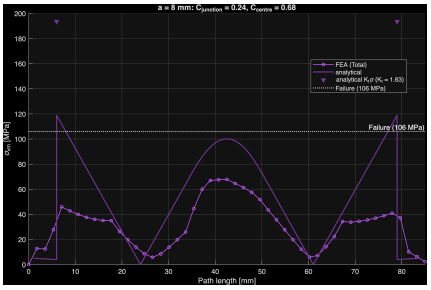
(a) $a = 2$



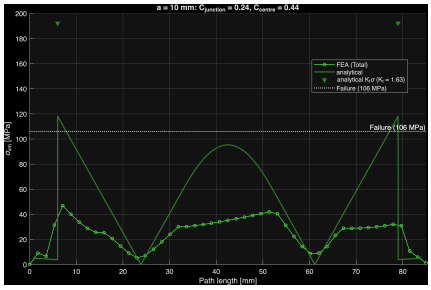
(b) $a = 4$



(c) $a = 6$



(d) $a = 8$



(e) $a = 10$

Figure 5: FEA (markers) and analytical (solid) σ_{vm} along the path for each patch radius (mm), at 400 N.

carries it more locally. As a grows, the FEA centre stress drops much faster than the beam predicts. This is the plate effect again: a wide patch is shared in two directions, which a one-dimensional strip cannot capture. The $a = 4$ mm case ($C_c = 0.33$) breaks the otherwise monotonic trend and should be re-checked, most likely for the path position relative to the patch or the mesh under it.

Junction. The analytical junction stress is three to six times the FEA value ($C_j = 0.15$–0.33). Several simplifications act in the same direction: (i) the grouser is assumed to clamp the skin rigidly, whereas the 5 mm block itself rotates and relieves the junction moment; (ii) Peterson's chart is for a symmetric shoulder, which is more severe than the one-sided T-junction here; (iii) the moment is averaged over the full width, which over-states the moment carried at the junction line far from the patch; (iv) the FEA path samples every 1.77 mm, so it can step over the fillet, which is only 1 mm long, and miss the true local peak. The overlays support (i) and (iv): the FEA rises towards the junction on the analytical slope but stops short of it, and peaks a few millimetres inside the span.

Implications. Even with these corrections, both models agree that a sharp rock ($a \leq 2$ mm) at 400 N drives the skin centre above the 106 MPa fatigue limit. Under repeated loading this location will initiate cracks before the junction.

6 Conclusions

A fixed stepped-beam model with a Peterson stress concentration reproduces the shape of the FEA stress distribution along the skin and, for sharp contacts, the centre peak to within about 10%. It is conservative at the junction by a factor of three to six, mainly because it treats the grouser as a rigid clamp and uses a symmetric-shoulder K_t. Sharp rocks ($a \approx 2\,\text{mm}$) push the centre stress above the fatigue limit. Tier 2 will refine the path sampling at the fillet, re-check the $a = 4\,\text{mm}$ case, and use the calibration factors to map safe designs against rock sharpness, fillet radius and skin thickness.

Repository resources

- Analytical model: `scripts and outputs/tier_1/path_stress_analytical.m`
- FEA comparison: `scripts and outputs/tier_1/fea_vs_analytical.m`
- Output figures and tables (with README): `scripts and outputs/tier_1/output`
- FEA path data: `Simulation results/Tier 1`
- CAD models: `CAD_models/Tier 1`
- Images used in this report: `media`

References

[1] NASA Jet Propulsion Laboratory, Mars Science Laboratory (Curiosity) mission reports and status updates on wheel wear. `https://mars.nasa.gov/msl/`

[2] Ansys Inc., *Ansys Mechanical User's Guide, Release 2025 R1*: Linearized Stress.

[3] Project repository, `github.com/Shass27/curiosity-perseverance-wheel-fea`